Does the LeHome $\pi_{0.5}$ teacher grasp? Per-grasp measurement in its original simulation (Run A)

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measurement run and write-up by Claude, 2026-10-19; v2, corrected the same evening (lift criterion, Section 5)

Abstract

We ask whether the $\pi_{0.5}$ LeHome teacher (`lehome_sim`) can grasp cloth, measured grasp by grasp rather than by fold scores. Run A puts the teacher in the simulation it was trained in (LeHome's shipped physics: a ∼150 kg sticky shirt under doubled gravity, round crossing jaws, LeHome's arm) and logs every jaw close on 20 episodes (4 garment types). Of 80 grasp attempts, 65 (81%) pinched cloth and 64 of those (98%, 95% CI 92–100%) lifted it at least 4 cm with the jaw (55 at least 6 cm; median rise 8.5 cm, median carry 15.7 cm); 18/20 folds (90%, CI 70–97%) passed LeHome's check. A typical grasp takes 3.0 layers. Our stricter criterion, at least half of the pinched vertices staying within 2 cm of the jaw-tip point until release, was met by only 2 grasps; it turns out to be unfit for this physics, where the ∼150 kg cloth sags out of that radius while still on the jaw. In its own simulation the teacher therefore pinches and lifts in almost every attempt, never one layer at a time. We list the ways this measurement can mislead and what Run B (corrected arm, cloth and contact) must show to distinguish "the teacher can grasp" from "it only works in its own physics".

1 Why measure grasps, not folds

Fold scores tell us whether a policy finished a fold, not which sub-skill failed. On 2026-10-02 the teacher passed 3/8 folds under corrected physics and most shirts shrank, which pointed at grasping, but no grasp had ever been scored. The project's plan is to RL only the sub-skills the teacher cannot do, so we need the grasp answer on its own. Run A is the control arm of an A/B test: the teacher in its own simulation, with every grasp labelled. Run B repeats the same 20 garments with the physics corrected to our measurements (arm 10% from reality, cloth 28%, contact calibrated).

2 Setup

Simulation. LeHome's Isaac Sim 5.1 environment as shipped (settings in Appendix B): particle cloth with 10^{-2} kg per vertex (a 14,746-vertex shirt weighs ∼147 kg), gravity scale 2, adhesion 0.1 (the cloth sticks to the jaws), LeHome's SO-101 model with its round capsule jaw colliders that cross at the tips, LeHome's actuator settings, 1/90 s physics. Tops start centred and unrotated on the table (our scene edit, the same in Run B); pants keep LeHome's randomised spawn.

Policy. Ilia Larchenko's `lehome_sim` $\pi_{0.5}$ checkpoint under his evaluation harness (`run_eval.py`, remote policy, chunked actions with inpainting, best-of-n candidates, retry on), with the speed patches we measured earlier (none touch physics or the model input). Two Isaac workers and one policy server on one NVIDIA L4.

Garments and episodes. The first five "Seen" garments of each type, one episode each, at most 600 policy steps, LeHome's own check-point success criterion. Workers took garments in queue order, so pants ran first. The whole run took 25m 20s on the L4 (76 s per episode with two workers).

3 The grasp benchmark

Every policy step, the logger (`oracle/grasp_log.py`) reads both grippers' joint angles and the cloth vertex positions.

- **Close / open.** A jaw "closes" when its joint drops below 0.19 rad after having been open (> 0.25 rad); it "opens" when it rises above 0.25 rad again.
- **Pinched set.** The vertices inside a 14 mm $\times$ 24 mm $\times$ 28 mm box between the pad faces of the jaw at the close, six steps after the close began (so the jaw has settled). Fewer than five vertices is an *empty* grasp.
- **Layers.** The pinched vertices are split into mesh-connected pieces; pieces of at least three vertices count as layers.
- **Held.** At every step while closed, the share of the pinched vertices still within 2 cm of the jaw tip. If that share ever drops below 50% the grasp is *slipped*, and the lift at that moment is recorded. Otherwise it is *held*, qualified by what the jaw did: *lifted* ($\geq$2 cm up), *dragged* ($\geq$3 cm sideways) or *static*.
- **Lift and carry.** Lift is the pinched vertices' median height above where they were at the close; carry is the jaw tip's travel while closed.
- **Parked closes.** A jaw closing on nothing at the end of the episode (the arm returning home) is not counted as an attempt.

The cloth reader takes the garment's pose live at every call. Isaac moves the garment prim to the cloth's bounding-box centre on every write-back, so a pose cached at the episode start drifts as soon as the cloth is lifted; an earlier version of the reader cached it, was caught before this run's data was collected, and was fixed (Section 5).

4 Results

Garment type	Episodes	Folds passed	Attempts	Pinched	Lifted $\geq$4 cm	Held (2 cm)	Kept %	Slip at (cm)	Max lift (cm)	Carry (cm)
Long-sleeve tops	5	5	25	20	20	0	12.5	1.9	10.6	21.4
Short-sleeve tops	5	5	30	20	19	0	8.5	3.5	8.1	10.1
Long pants	5	3	15	15	15	0	7.0	2.3	6.7	13.8
Shorts	5	5	10	10	10	2	18.0	5.1	8.0	16.4
All	20	18	80	65	64	2	8.0	2.7	8.5	15.7

Table 1: Run A by garment type. Pinched: attempts that caught at least five vertices. Lifted: pinched grasps whose cloth rose at least 4 cm with the jaw. Held: pinched grasps that kept at least half of those vertices within 2 cm of the jaw-tip point until it opened. Kept, slip, lift and carry are medians over pinched grasps.

The teacher pinches and lifts. Most attempts catch cloth (65/80; the 15 empties are all on tops, Section 5), and the pinched cloth then rises with the jaw: 64 of 65 grasps lifted it at least 4 cm, 55 at least 6 cm (median rise 8.5 cm, Figure 3), with 15.7 cm of travel while closed. The 2 cm "held" criterion tells a different story (Figure 2): the median grasp kept only 8.0% of its pinched vertices within 2 cm of the tip point, falling below 50% after 2.7 cm of lift, and only 2 grasps met it. Both are true at once because the cloth is $\sim$150 kg: the pinched patch sags several centimetres below the tip point while hanging from the jaw. The lift numbers are the grasp result; the 2 cm numbers measure sag.

Grasps take several layers. The median grasp caught 3.0 mesh-connected layers (Figure 4); 2 of 65 caught a single layer. On pants the jaw closes over waistband and leg layers at once; on tops it takes the front and back of a sleeve or hem together.

Folds pass anyway. 18 of 20 episodes passed LeHome's check (the harness's own summary agrees: 18/20, SR 90% [79.0%-100.0%]). Episodes that passed had a mean kept share of 16.1% against 9.5% for those that failed (too few failed folds for a test; Figure 5): the grasp quality we measure does not separate passed from failed folds in this physics. Table 2 lists every episode.

5 What could have gone wrong

1. **The 2 cm "held" criterion does not fit this physics (the correction behind v2 of this report).** It was designed for flat pads and a light cloth, where a held patch stays at the pads. LeHome's cloth weighs $\sim$150 kg, so a patch hanging from the round jaws sags out of the 2 cm radius while still attached; the logger's independent lift measure

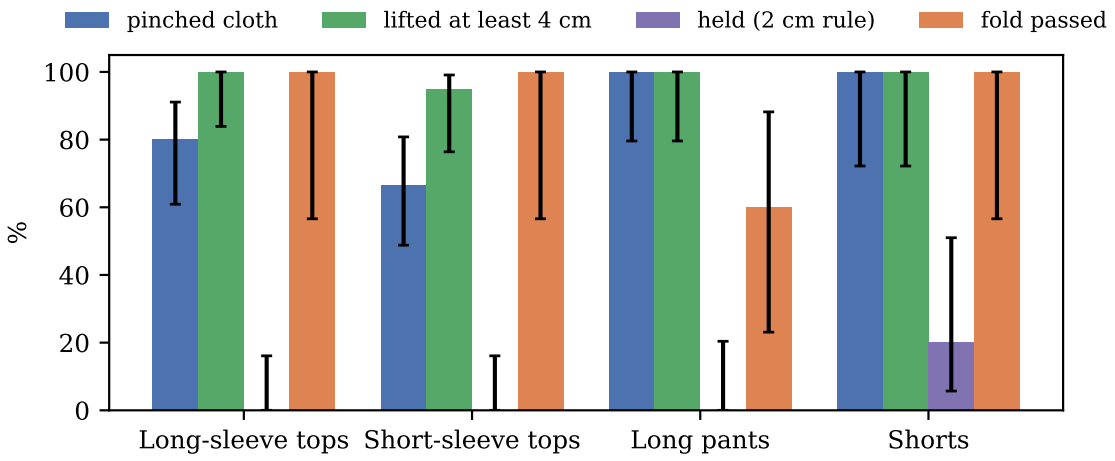


Figure 1: Per type: the share of attempts that pinched cloth, the share of pinched grasps whose cloth rose at least 4 cm with the jaw, the share that met the 2 cm rule, and the share of folds that passed, with 95% Wilson intervals. With five episodes per type the fold intervals are wide.

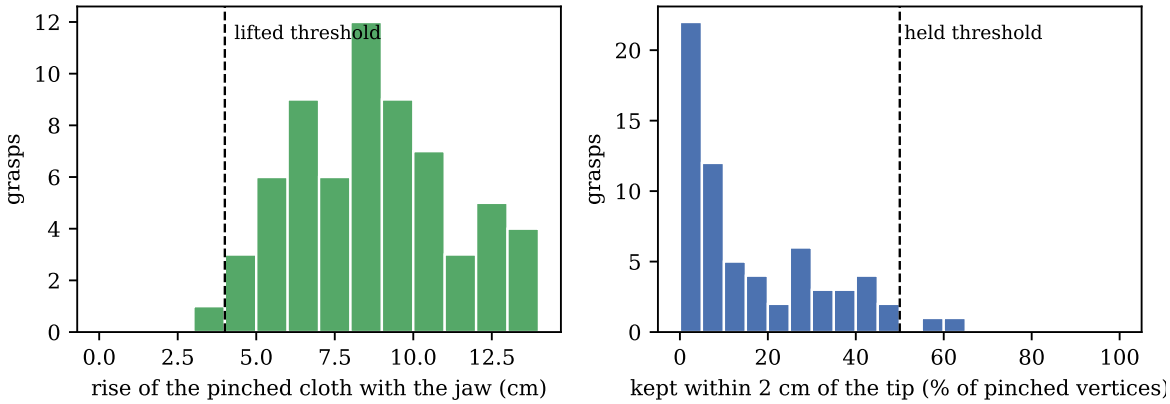


Figure 2: Left: how far the pinched cloth rose with the jaw (the grasp result). Right: the share of the pinched vertices that stayed within 2 cm of the tip point until release (the sag). All pinched grasps; dashed lines are the thresholds.

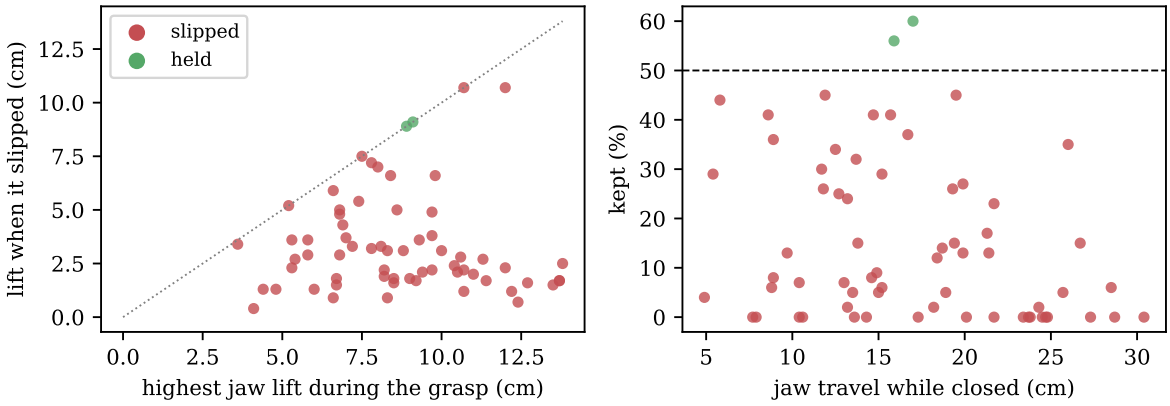


Figure 3: Left: the lift at which a grasp slipped against the highest lift the jaw reached in that grasp (dotted line: equal). Points far below the line slipped early and the jaw kept rising. Right: cloth kept against jaw travel while closed.

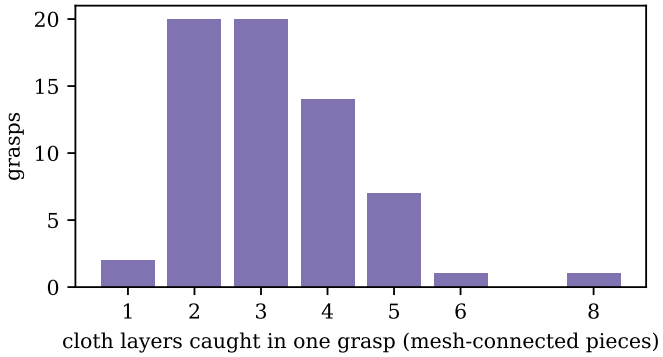


Figure 4: Layers caught per pinched grasp.

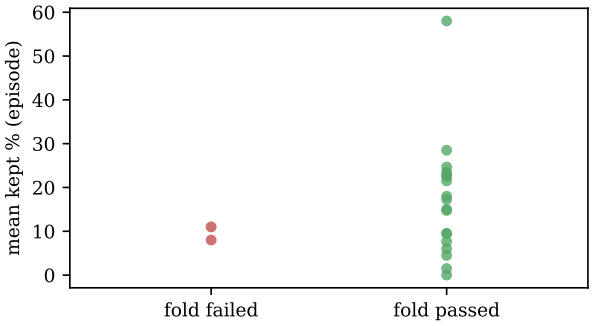


Figure 5: Mean kept share per episode against the fold result.

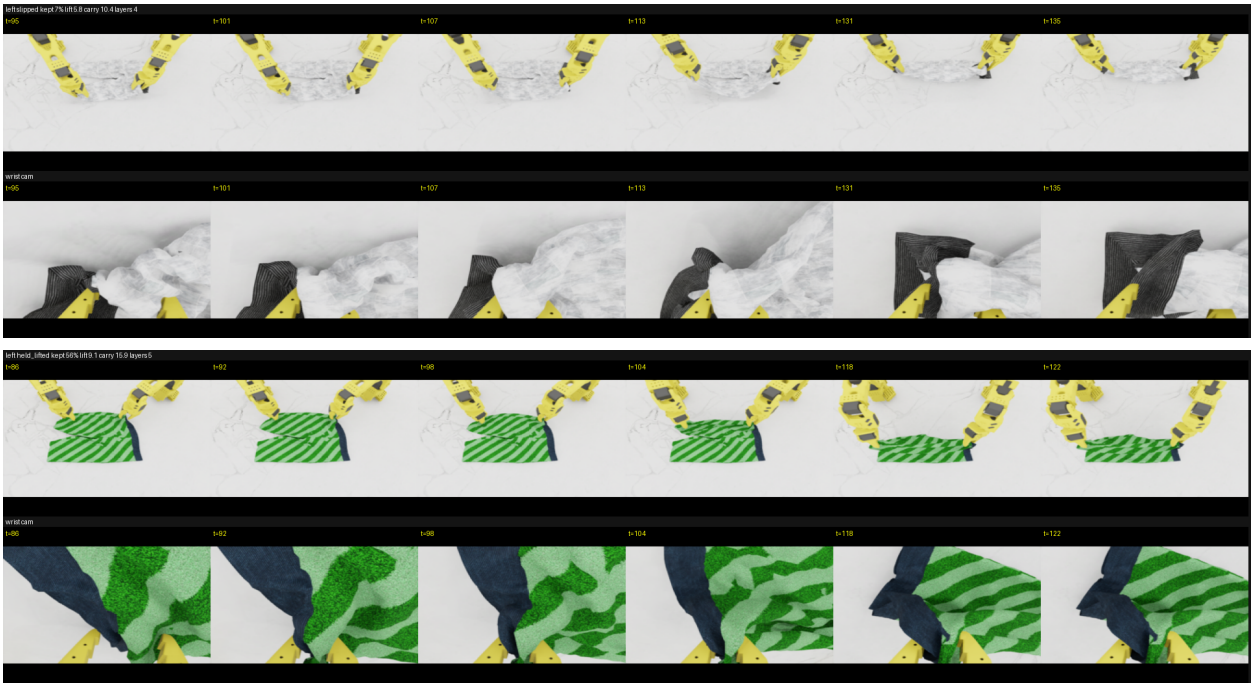


Figure 6: What the labels look like from the cameras, one grasp per block: top camera (row 1) and the grasping arm's wrist camera (row 2) six steps before the close, at the close, when the logger declared the slip, twelve steps after the close, and just before and after the release. Upper block: a slipped-labelled long-pants grasp (kept 7%, max lift 5.8 cm, carry 10.4 cm, 4 layers): by the logger, 93% of the pinched vertices were more than 2 cm from the tip point at $t = 107$, yet in the wrist camera cloth is still bunched at the jaw at $t = 113$. The garment did move with the jaw and the fold passed. This is the ambiguity of Section 5, item 1: a strict vertex-level criterion against round jaws that drag a sticky, very heavy cloth. Lower block: a held-labelled shorts grasp (kept 58%, lift 9.1 cm).

shows the cloth rising with the jaw in 64 of 65 grasps. The first version of this report read the 2 cm numbers as "slipped" and called the grasps partial drags; that was wrong for this physics. What the lift measure cannot tell is whether the cloth was still up at the moment of release (added to the logger as `lift_open_cm` for the next runs). The same jaw geometry can also explain the 15 empty attempts, all on tops: the jaw closed at the cloth but the pad box held fewer than five vertices. In Run B's light cloth with flat pads the two criteria should agree.

2. **Grasps that drag still move cloth.** In a sticky, very heavy cloth, dragging a small patch moves the whole garment, so a grasp that keeps 7% of its vertices can still complete a fold. This is a property of the training physics, not of the policy, and it is exactly what Run B removes.

3. **The teacher's strategy is sliding, not pinching.** The jaw mostly lands flat on the cloth and slides, taking several layers. That worked here; whether it works when a grasp must actually hold is Run B's question.

4. **Reader bug, fixed before the data.** The first launch of Run A used a cloth reader that cached the garment pose; it was stopped after five minutes, fixed, and relaunched. All data here come from the relaunch. The same bug had corrupted our first scripted contact numbers that morning; those were re-measured.

5. **Sample size.** Five episodes per type give wide intervals on fold rates (Figure 1); grasp-level rates (65 pinched grasps) are tighter. Differences between types should not be read beyond the intervals.

6. **Spawn asymmetry.** Tops start centred and unrotated, pants randomised; both runs share this, so it does not bias A against B, but it does make the type rows not strictly comparable.

7. **Parked closes.** 13 of the 93 jaw closes were the jaw shutting on air while returning home; they are excluded from attempts. Had they been counted, the empty share would read 30% rather than 19%.

8. **The harness is on the teacher's side.** Retry, inpainting and best-of-n candidates are part of Ilia's policy and stay on in both runs. They raise fold success; they do not change what a single grasp does.

6 What to look for in Run B

Run B uses the same 20 garments and policy with: our fitted arm (10% from the real SO-101 on held-out motions), the measured cloth (28% from a real hanging sheet, on flattened garment models so the stiff settings lie flat), flat pads, and a grasp model in which cloth caught between the pads moves with the jaw and lets go above $2\mu N$ with the measured PLA–cotton friction and 9 N squeeze (Isaac's own contact is the fallback: it holds about half a sleeve through a carry and lets go 2–10$\times$ too early). The pre-registered reading, on the share of pinched grasps that hold:

- **Held $\geq$ 80% in B and empty $\leq$ 20%:** the teacher can grasp once the physics is right; its failures are placement and sequencing, so search and SGS go there.
- **Held in A but $<$ 50% in B:** physics mismatch; a small fine-tune of the teacher in the corrected physics.
- **Held $<$ 50% in both:** grasping itself is the gap; it needs SGS-style RL.

On the 2 cm criterion Run A has held $= 3\%$; on the lift criterion it has 98%, which changes which branch Run B can land in (see below). Beyond the rule, watch (i) whether the pinched share stays near 100% with flat pads and light cloth (the jaw must now land precisely), (ii) whether kept shares jump once cloth is actually held, (iii) how many layers the flat pads take, and (iv) whether folds pass when a grasp that keeps 7% of its vertices no longer moves the garment.

Because the 2 cm criterion failed in Run A's physics, read Run B primarily on the lift criterion (pinched cloth rises $\geq$4 cm with the jaw and is still up at release), with the 2 cm criterion as the stricter secondary; Run A's lift share is 98%, so on that criterion the teacher does grasp in its own world, and the open question is only whether the same motions grasp under real contact.

A Per-episode results

Episode	Fold	Attempts	Pinched	Held	Kept %	Max lift	Layers
Top_Long_Seen_0	pass	5	4	0	9.5	11.5	2 2 2 3
Top_Long_Seen_1	pass	5	4	0	12.5	10.3	2 2 3 3
Top_Long_Seen_2	pass	5	4	0	2.5	9.9	2 2 2 3
Top_Long_Seen_3	pass	5	4	0	26.5	10.1	2 2 3 3
Top_Long_Seen_4	pass	5	4	0	7.5	12.6	2 2 2 2
Top_Short_Seen_0	pass	6	4	0	18.5	9.7	2 3 3 6
Top_Short_Seen_1	pass	6	4	0	21.0	7.8	1 2 2 3
Top_Short_Seen_2	pass	6	4	0	11.0	9.8	3 3 3 3
Top_Short_Seen_3	pass	6	4	0	1.0	6.2	3 4 4 4
Top_Short_Seen_4	pass	6	4	0	35.0	4.8	2 4 4 4
Pant_Long_Seen_0	fail	3	3	0	7.0	6.7	3 4 4
Pant_Long_Seen_1	pass	3	3	0	8.0	6.6	3 4 4
Pant_Long_Seen_2	pass	3	3	0	6.0	5.8	3 5 5
Pant_Long_Seen_3	fail	3	3	0	0.0	6.8	3 4 8
Pant_Long_Seen_4	pass	3	3	0	25.0	8.3	4 5 5
Pant_Short_Seen_0	pass	2	2	0	21.5	8.4	2 4
Pant_Short_Seen_1	pass	2	2	0	0.0	7.3	2 5
Pant_Short_Seen_2	pass	2	2	0	22.5	10.1	1 5
Pant_Short_Seen_3	pass	2	2	2	58.0	9.0	3 5
Pant_Short_Seen_4	pass	2	2	0	18.0	6.1	3 4

Table 2: Every episode of Run A. Kept and max lift are medians over the episode's pinched grasps; layers lists the layers of each pinched grasp.

B Grasp-benchmark parameters

Held radius 2 cm; slip threshold 50% kept; minimum pinch 5 vertices; lift threshold 2 cm; drag threshold 3 cm; minimum layer 3 vertices; close below 0.19 rad, open above 0.25 rad (jaw open 0.55, closed -0.17); pad box $x \in [-10, 4]$ mm, $|y| < 12$ mm, $z \in [-103, -75]$ mm in the gripper-link frame.

C Physics settings of this run

Setting	Value
particle_mass	0.01
stretch_stiffness	1e+08
bend_stiffness	5000
shear_stiffness	5000
spring_damping	10
gravity_scale	2
adhesion	0.1
friction	0.5
particle_friction_scale	0.6
solver_position_iteration_count	32

With 10^{-2} kg per vertex the 14,746-vertex shirt weighs $\sim$147 kg, about 3,000 times a real one; gravity scale 2 doubles its weight again; adhesion 0.1 makes the cloth stick to whatever touches it.